

KARNATAKA BOARD

CLASS 10

SCIENCE

2015

Max. Marks: 80

General Instructions to the Candidate:

- i) The Question-cum-Answer Booklet consists of objective and subjective types of questions having 55 questions.
- ii) This question-cum-answer booklet contains two Parts. Part – A contains the questions of Physics and Chemistry and Part – B contains Biology questions.
- iii) The question-cum-answer booklet has 36 questions in Part – A and 19 questions in Part – B.
- iv) Space has been provided against each objective type question. You have to choose the correct choice and write the complete answer along with its letter in the space provided.
- iv) For subjective type questions enough space for each question has been provided. You have to answer the questions in the space.
- v) Follow the instructions given against both the objective and subjective types of questions.
- vi) Candidate should not write the answer with pencil. Answers written in pencil will not be evaluated. (Except Graphs, Diagrams & Maps).
- vii) In case of Multiple Choice, Fill in the blanks and Matching questions, scratching / rewriting / marking is not permitted, thereby rendering to disqualification for evaluation.
- ix) Space for Rough Work has been printed and provided at the bottom of each page.
- x) Candidates have extra 15 minutes for reading the question paper. xi) Do not write anything in the space provided in the right side margin.

Four alternatives are given for each of the following questions / incomplete Statements. Only one of them is correct or most appropriate. Choose the correct alternative and write the complete answer along with its letter of alphabet

1. Electronic configuration of silicon is

- (A) $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$ (B) $1s^2, 2s^2, 2p^6, 3s^2, 3p^3$
(C) $1s^2, 2s^2, 2p^6, 3s^2$ (D) $1s^2, 2s^2, 2p^6, 3s^2, 3p^4$.

Ans: (A) $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$

2. A cyclist bends towards the centre while going in a circular path to

- (A) Move the cycle slowly
(B) Gain necessary centrifugal force
(C) Gain necessary centripetal force
(D) Increase the friction.

Ans: (C) gain necessary centripetal force

3. Which of the following is not a form of solar energy?

- (A) Fossil fuel energy (B) Wind energy
(C) Hydro energy (D) Nuclear energy.

Ans: D) Nuclear energy.

4. The alloy that contains copper and zinc as its constituents, is

- (A) Brass (B) bronze
(C) German silver (D) gunmetal.

Ans: (A) brass

5. The measure to reduce energy crisis is

- (A) Wastage of water
(B) Using fluorescent tubelights
(C) Luxurious life style
(D) Each individual uses his own vehicle

Ans: (B) Using fluorescent tube lights

6. The reaction that is considered to be the source of solar energy is

- (A) radioactivity
- (B) Thermonuclear fusion reaction
- (C) Nuclear fission reaction
- (D) Chemical reaction.

Ans: (C) nuclear fission reaction

7. The reducing agent used in the extraction of silicon is

- (A) Magnesium (B) Silica
- (C) Phosphorus (D) Sulphur.

Ans: (A) Magnesium

8. The apparent change in frequency of wave motion due to the relative motion between the source and the observer is known as

- (A) Raman effect (B) Doppler effect
- (C) Electromagnetic effect (D) Tyndall effect.

Ans: (B) Doppler effect

9. An example for a pentavalent dopant is

- (A) Boron (B) Antimony
- (C) Gallium (D) Indium.

Ans: (B) Antimony

10. The alkane among the following is

- (A) C_2H_2 (B) C_2H_4
- (C) C_6H_6 (D) C_2H_6 .

Ans: (D) C_2H_6

Fill in the blanks:

[3 × 1 = 3]

11. The device that converts solar energy into electrical energy is..... .

Ans: The device that converts solar energy into electrical energy is solar cells.

12. The lightly doped region of a transistor is

Ans: The lightly doped region of a transistor is base region.

13. The device used by the traffic control authorities to detect vehicle crossing speed limit is

Ans: The device used by the traffic control authorities to detect vehicle crossing speed limit is Speed Radar.

14. Match the names of hydrocarbons given in Column-A with their molecular formulae given in Column-B. Write the correct answer in the space provided: [4 × 1 = 4]

Column-A

Column-B

(a) Benzene

(i) C₄H₁₀

(b) Propyne

(ii) C₃H₄

(c) Butene

(iii) C₃H₆

(d) Butane

(iv) C₆H₆

(v) C₄H₈

(vi) C₄H₆

(vii) C₂H₂.

Ans:

(a) Benzene

C₆H₆

(b) Propyne

C₄H₆

(c) Butene

C₄H₈

(d) Butane

C₄H₁₀

Answer the following questions: $6 \times 1 = 6$

15. Write the balanced chemical equation for the following chemical reaction:

Magnesium reacts with dilute hydrochloric acid.

Ans: $\text{Mg(s)} + 2 \text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

16. What is cracking?

Ans: Cracking, in petroleum refining, the process by which heavy hydrocarbon molecules are broken up into lighter molecules by means of heat and usually pressure and sometimes catalysts.

17. Define half-life period of a radioactive element.

Ans: Time taken for a radioactive substance to decay to half of its original value. Decay means it loses energy or mass by emitting radiation. So if it started out with a certain mass, after some time it has half that mass; that is its half-life.

18. What is uniform circular motion?

Ans; When a body moves in a circular path with uniform speed or constant speed, its motion is known as Uniform Circular motion.

19. Write the uses of a diode.

Ans: The uses of a diode are following:

- Rectifying a voltage, such as turning AC into DC voltages
- Isolating signals from a supply
- Controlling the size of a signal
- Mixing signals

20. Name the system used to represent the brightness of a star.

Ans: Brightness of stars, apparent and absolute magnitude is measured by convention, taking another star as a standard. The scale called "apparent magnitude" is the usual measure.

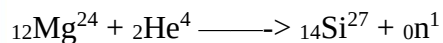
Answer the following questions: $9 \times 2 = 18$

21. What is induced radioactivity? Explain with an example.

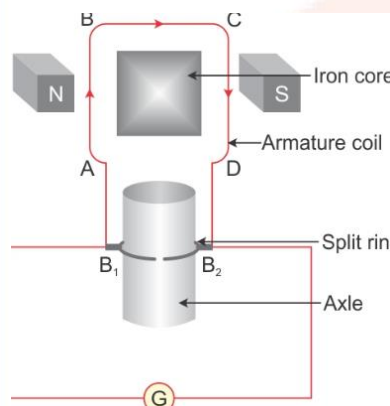
Induced Radioactivity is a type of Artificial Radioactivity. A process in which a new radioactive isotope is prepared by using a known element is known as Induced Radioactivity.

For Example:

In magnesium, the change is represented as:



22. Draw the diagram of a D.C. dynamo & label the parts.



DC DYNAMO

23. What are propellants? How do they work in vacuum?

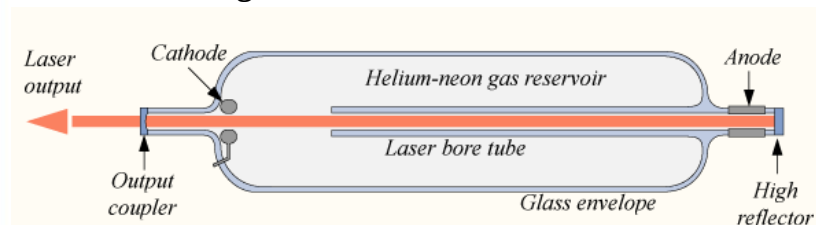
A propellant is a chemical substance used in the production of energy or pressurized gas that is subsequently used to create movement of a fluid or to generate propulsion of a vehicle, projectile, or other object. Common propellants are energetic materials and consist of a fuel like gasoline, jet fuel, rocket fuel, and an oxidizer. Propellants are burned or otherwise decomposed to produce the propellant gas. Other propellants are simply liquids that can readily be vaporized.

24. Write any four methods of conservation of water.

Ans: Four methods of conservation of water are followings:

- Protection of Water from Pollution.
- Redistribution of Water.
- Rational Use of Groundwater.
- Use of Modern Irrigation Methods.

25. Draw the diagram of Helium-Neon laser tube and label the parts.

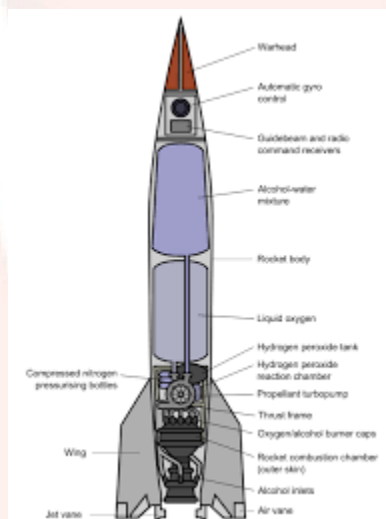


26. Write the differences between A.C. dynamo and D.C. dynamo.

Ans; Ac dynamo has full rings whereas Dc dynamo has split rings.

Ac dynamo produces alternating current whereas Dc dynamo produces direct current.

27. Draw the diagram of a single stage rocket and label the parts.



28. “Soaps are eco-friendly than detergents.” Justify this statement.

Soaps are natural, less harmful to us and environment, and are biodegradable. As such they do not cause harmful pollution to our rivers and other water bodies. Detergents are that they are not so environmentally friendly.

29. Laser is used in eye surgery and industries. State the property of laser related to these uses.

Lasers is used in the science of medicine. They are able to heal or otherwise improve people's physical well-being. There are number of uses for medical lasers. They are followings.

- Cutting of tissue in surgical procedures.
- Reshaping the cornea of the eye to improve sight.
- Cleaning clogged arteries; burning away cavities and whitening the teeth;

- Lasers are used in industry in a huge variety of applications. Materials processing includes cutting, drilling, welding, etc., and generally involves the use of high-powered lasers. Sing fiber optics, access to previously inaccessible locations is available.
- Laser processing is easily automated to allow computer and robot control.
- New types of processing has and continues to be developed, including producing new surface alloys and hardening of materials.

Answer the following questions: $4 \times 3 = 12$

30. What is polymerization? Name one example each of the two types of synthetic polymers.

Ans: Polymerization is a process in which relatively small molecules, called monomers, combine chemically to produce a very large chainlike or network molecule, called a polymer. The monomer molecules may be all alike, or they may represent two, three, or more different compounds. Usually at least 100 monomer molecules must be combined to make a product that has certain unique physical properties. Two classes of polymerization usually are condensation polymerization, and addition polymerization.

Example of synthetic condensation polymerization- Polyester and Polyamides.

Example of synthetic addition polymerization is Polystyrenes and Polyamides.

31. What is Raman Effect? Write the differences between Raman Effect and Rayleigh scattering.

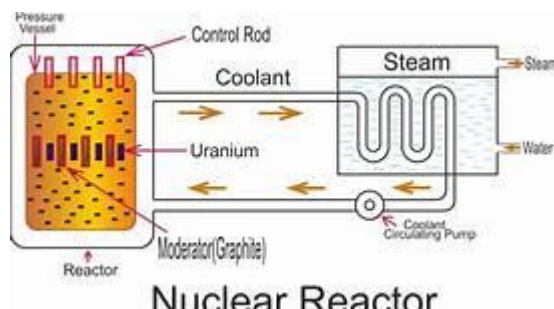
Ans: The phenomenon of scattering of light by a substance when the frequencies of radiations scattered at right angles are different generally lower and only occasionally higher from the frequency of the incident light, is known as Raman Scattering or Raman Effect.

Raman scattering is inelastic scattering from molecules. The photon interacts with the molecule and changes the molecules vibrational, rotational or electron energy.

Rayleigh scattering is in the main elastic scattering from small particles whose size is less than that of the wavelength of the photon. The scattering can occur of atoms or molecules and for molecules the scattering can be inelastic with a change of rotational energy of the molecule.

32. Draw the diagram of a nuclear reactor and label the following parts:

Moderator (b) Heat exchanger.



(a) Nuclear Reactor

33. (a) State Universal law of gravitation.

Universal law of gravitation states that every object in the universe attracts every other object with a force directed along the line of centres for the two objects that is proportional to the product of their masses and inversely proportional to the square of the separation between the two objects.”

(b) The distance between the two bodies is increased by two times.

Explain with the help of mathematical formula of gravitational law, whether the gravitational force between two bodies increases or decreases and by how many times.

Ans: Formula of gravitational law

$$F = G \frac{m_1 m_2}{d^2}$$

According to gravitational law the gravitational force decreases by $\frac{1}{4}$ times the original gravitational force between two bodies if the distance between the two bodies is increased by two times.

Answer the following questions:

$$3 \times 4 = 12$$

34. (a) What are galaxies ? Name three types of galaxies.

Ans: Galaxy is a huge collection of gas, dust, and billions of stars and their solar systems. A galaxy is held together by gravity. Our galaxy is the Milky Way. When we look up at stars in the night sky, we see other stars in the Milky Way. If it's really dark, far away from lights from cities and houses, we can even see the dusty bands of the Milky Way stretch across the sky. There are three main types of galaxies: Elliptical, Spiral, and Irregular.

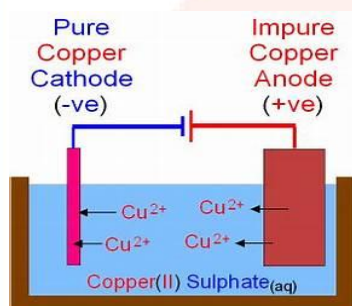
(b) Why do the sun spots appear dark in colour?

Ans: Even though sunspots are fiery white hot, they are somewhat cooler than the part of the Sun around them. When you inspect the Sun through a very dark filter which is the only safe way, the Sun itself does not look very bright, and sunspots appear dark in comparison.

35. Draw the diagram of an electrolytic cell used in the extraction of copper and label the following parts:

(a) Cathode (b) Anode (c) Copper sulphate solution.

Ans:



ELECTROLYSIS OF COPPER

36. (a) Write the formula to calculate the efficiency of a heat engine.

Ans: The efficiency of a heat engine

$$\text{Efficiency} = W_{\text{net}}/Q_h = Q_h - Q_c / Q_h$$

$$= 1 - Q_c/Q_h$$

$$= W_{\text{net}} = \text{Net work done by engine.}$$

$$Q_h = \text{Energy added as heat}$$

$$Q_c = \text{Energy removed as heat.}$$

(b) Mention any three advantages of an internal combustion engine.

Ans: An internal combustion engine or petrol engine, is smaller and lighter.

An internal combustion engine or petrol engine can be started at once at a moment's notice.

An internal combustion engine or petrol engine is quite safe to use.

PART - B (Biology)

Four alternatives are given for each of the following questions / incomplete statements. Only one of them is correct or most appropriate. Choose the correct alternative and write the complete answer along with its letter in the space provided against each question.

$5 \times 1 = 5$

37. is commonly called personality hormone.

(A) Adrenaline (B) Thyroxine

(C) Insulin (D) Glucagon.

Ans: (B) Thyroxine

38. Which of the following vertebrates do not have teeth?

(A) Amphibia (B) Aves

(C) Mammals (D) Reptiles.

Ans: (D) Aves

39. The red pigment present in Polysiphonia is

(A) Phycoerythrin (B) Phycocyanin

(C) Chlorophyll (D) Xanthophyll.

Ans: (A) Phycoerythrin

40. The number of chambers in the heart of fish are

(A) four (B) three

(C) two (D) one.

Ans: (C) two

41. The processes involved in oxygen cycle are

(A) Nitrification, Respiration

(B) Nitrification, Denitrification

(C) Photosynthesis, Denitrification

(D) Respiration, Photosynthesis.

Ans: Respiration, Photosynthesis

42. Match the names of epithelial tissues given in Column-A with their structure and location given in Column-B. Write the correct answer in the space provided: $4 \times 1 = 4$

Column-A

Column-B

(a) Squamous epithelium

(i) elongated cells, larynx

(b) Columnar epithelium

(ii) ciliated cells, small intestine

(c) Ciliated epithelium

(iii) elongated cells, sweat gland

(d) Cuboidal epithelium

(iv) ciliated cells, sweat gland

(v) flat cells, oesophagus

(vi) cube shaped cells, salivary gland

(vii) flat cells, small intestine

Ans:

Column-A

Column-B

Squamous Epithelium

flat cells, oesophagus

Columnar epithelium

ciliated cells, small intestine

Ciliated epithelium

ciliated cells, sweat gland

Cuboidal epithelium

cube shaped cells, salivary gland

Answer the following in a sentence each:

$4 \times 1 = 4$

43. It is hard to detect the colour of objects during night in dim light. Give reason.

Ans: The retina contains two different types of receptor cells which are called cones and rods. Cones are used to detect colour and rods are used to detect the brightness or dimness. The cone cells are more sensitive than rod cells in bright light. Because of this, it is more difficult to detect the colours in dim light.

44. What is food adulteration?

Ans: Food Adulteration is the process by which the quality or the nature of a given food is reduced through addition of adulterants or removal of vital substance. Foreign and usually

inferior chemical substance present in food that cause harm or is unwanted in the food. Basically, during food adulteration, small quantity of non-nutritious substances is added intentionally to improve the appearance, texture or storage properties of the food.

45. Human pinna is flexible. Why?

Human pinna is flexible because it is found in places where we need some support and structure, but a bit of flexibility as well. This includes places such as our joints, our ears, and our nose, as well as in between the vertebrae in our spinal column.

46. Fibres are used to make gunny bags. Why?

Jute is used to make gunny bags because jute is a very strong material.

Answer the following questions in two or three sentences each: $6 \times 2 = 12$

47. Write any two characteristic features of reptiles.

- Most Reptiles Lay Eggs- Reptiles are amniote animals, which mean that the eggs, laid by females, contain an elastic sac within which the embryo develops.
- The Skin of Reptiles Is Covered With Scales (or Scutes)-

The scales of reptiles, which develop from the epidermis, the outermost layer of skin, are small, hard plates made of the protein keratin

48. What are the observations to be noted while purchasing packed fruit juice?

The observations to be noted while purchasing packed fruit juice are expiry date, nutritional value and whether the pack is puffed.

49. Forests help to reduce global warming. Justify.

Ans: Trees in forest help reduce global warming by taking carbon dioxide from atmosphere and by giving oxygen. Trees absorb carbon dioxide and it reduces the level of carbon dioxide in atmosphere, which actually forms a blanket in the atmosphere trapping all the infrared rays and heat causing global warming.

50. Mention any two disadvantages of using genetically modified foods.

Ans: Genetically modified foods are created by inserting foreign genes into an organism. This process has adverse effects on humans because these inserted genes may carry along with them specific allergens that trigger such an immune response.

Genetically modified foods are void of nutritional value. As genetic engineering tends to focus more on increasing their production, prolonging their lifespan and ability to deter pest, the nutritional value of some crops are comprised.

51. Write any two differences between reservoir pool and exchange pool.

Reservoir pool	Exchange pool.
1. It is the primary source of the element or the nutrient storehouse of the chemical.	1. It is characterized by Rapid exchange of element.
2. In this pool chemical element is locked or retained for a long period of time	2. Exchange pool holds the element for a short period of time. and is represented by the biotic factors which come in minor part of abiotic face
Generally reservoirs are large and are abiotic factors.	Generally exchange policy small It is represented by the biotic factors which come in minor part of abiotic phase.

52. A monocot leaf is more vulnerable to wear and tear than a dicot leaf. Why?

A monocot leaf is more vulnerable to wear and tear than a dicot leaf because monocots leaves mostly have parallel veins whereas, dicot leaves have net-veins.

Answer the following questions:

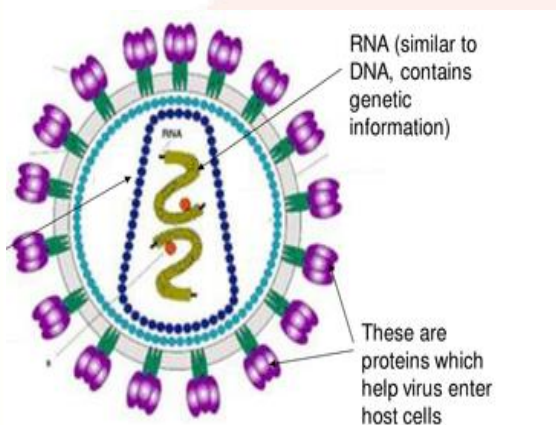
$$2 \times 3 = 6$$

53. Explain the functioning of human ear.

Ans: The major functions of the ear are to maintain the balance and equilibrium of the body and to aid in hearing capabilities. For example, the ear has three main portions, each having its own

role to play. The outer ear is the pinna, and its function is to gather the sound waves like a funnel and transmit to the middle ear through the ear canal. Different parts of the ear and their functions are very important for hearing. The vestibular window separates the middle ear from the inner ear. The inner ear is also termed as cochlea, which is a dual membrane delicate structure, with fluids filled between the membranes and lined with hair cells. The movement of the hair cells that line cochlea, in response to the sound waves, is like that of grasses in the meadows. It serves to send electrical impulses to the brain, i.e. the master organ that accurately identifies the sounds we hear.

54. Draw a diagram showing the structure of HIV and label any two parts.



STRUCTURE OF HIV

55. Draw a diagram showing the vertical section of human eyeball and label the following:
4

(a) Lens (b) Optic nerve

